LAST NAME: _____

Practice Quiz 08D — MATH 2055 — Fall 2026

Practice for Quiz 08

The following is closed book and closed note. Don't assume the grader will know what you mean. Use full sentences and provide complete and correct solutions.

Revised using GPT-6 Astra in the Codex harness.

1. Let $F : \mathbf{R} \to [2, \infty)$ be given by $F(x) = x^2 + 2$.

   (a) Show that F is not one-to-one.

   (b) Choose a subset $D \subseteq \mathbf{R}$ such that the restriction $F : D \to [2, \infty)$ has an inverse. Give the inverse and verify both compositions, with their domains stated.